

Integrating hyperbolic trig functions

Key Points

For integration recall the differentiation facts. Common ones:

$$\int \cosh x \, dx = \sinh x$$

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$$\int \operatorname{sech}^2 x \, dx = \tanh x$$

An alternative is to use the exponential form or identities or formula sheet

Basic Skills

Q1. Find $\frac{d}{dx}(2\sinh 4x)$

Q2. Find the gradient of $y = \tanh x$ at $x = 0$

Q3. State $\int e^{kx} \, dx$

Q4. Use the chain rule to find $\frac{d}{dx}(\cosh(4 - x^2))$

Q5. Find $\int \sinh 2x \, dx$

Q6. Explain why $\int \cosh x \, dx = \sinh x$

Q7. Using exponential form prove that $\int \sinh x \, dx = \cosh x$

Q8. Evaluate $\int_1^2 \operatorname{sech}^2 x \, dx$

Competency Skills

Q9. Find the following integrals

a) $\int 2 \cosh 5x \, dx$

b) $\int x \sinh 2x \, dx$

c) $\int x \operatorname{sech}^2(x^2 + 1) \, dx$

Q10. Using integration by parts integrate the function $f(x) = \operatorname{arccosh} x$

Q11. Evaluate $\int_0^2 \operatorname{arcsinh} x \, dx$

Q12. Integrate $\operatorname{cosech} x$ using exponential form with substitution $u = e^x$

Q13. Find the area bound by the curve $y = \cosh^2 x$ and the lines $x = -2, x = 2, y = 0$

Advanced Skills

Q14. For the function $f(x) = \coth x$

a) Find an expression for the integral $\int_a^b f(x) \, dx$

b) Hence determine if the integral $\int_{-2}^2 f(x) \, dx$ exists and if so calculate it's value

c) Hence determine if the area bound by the function past $x = 1$ exists and if so calculate it's value

Q15. Find the anti-derivative of $x^3 \cosh x^2$

Q16. Find the exact area of the region in the positive, positive quadrant bound by the inverse hyperbolic trig functions $y = \operatorname{arccosh} 2x$ and $y = \operatorname{arcsinh} x$

Extension

Q13. Using the derivatives of inverse hyperbolic trig functions state some integral rules